

Active v0.3.0 - frozen at b8d8d93; reserved evaluation complete, public asset publication pending

Task: real-versus-AI classification with a label, AI-positive score, CPU web/API/batch interface and model-influence explanation. Public-data results only; organizer data, baseline and hidden scores were not supplied.

Model, data and operating point

Frozen generic **CLIP ViT-L/14** image tower plus our trained logistic head.
Training: **8,000 CIFAKE + 6,239 GenImage BigGAN/SD1.5** images; original and identical crop/JPEG views, 90% GenImage loss weight. C=10 selected on internal validation. The pretrained tower is credited to OpenAI; we trained the head.
Inference: EXIF orientation, official 224px CLIP centre crop, L2-normalized embedding, linear logit and temperature scaling. Temperature **0.946907**; AI threshold **0.821428**. Temperature uses separate calibration data; threshold uses internal validation, not external labels.
Calibration: 9,969 CIFAKE + 781 GenImage; validation sizes below. Exact/perceptual overlap screening and grouped splits are heuristic. Backbone pretraining overlap is unknown. GenImage-derived head: noncommercial research use.

Runnable interface and attribution

Shared CPU CLI/API; single image or batch with one loaded model. JPEG/PNG/WebP uploads up to **25 MiB / 40 MP**. EXIF is separate and does not alter predictions; C2PA presence is a heuristic substring scan, not JUMBF parsing or signature verification.
Tower **608,352,029 bytes** + head **7,949 bytes**, SHA-256 verified.
CPU-only model runtime: torch, without clip/torchvision/ftfy/regex; ordinary app dependencies remain required. Public assets and clean-download setup timing are pending.
Measured CPU components: predict **~0.38s/image** (ResNet ~64ms), explanation **~5.5s**, robustness **~5.1s**. Not combined-request guarantees. Input-gradient influence map uses a 14px patch grid plus masking diagnostic.

Measured development performance at the stated operating point

Evaluation	AUC	Macro-F1	Accuracy	Real FPR	AI recall
GenImage validation (783)	0.991	0.950	95.0%	1.9%	92.3%
CIFAKE validation subset (4,000)	0.910	0.759	76.7%	5.0%	58.4%
External dev - original	0.771	0.604	63.5%	11.7%	38.8%
External dev - format matched	0.789	0.608	64.2%	10.0%	38.5%

External rows average two generator/source pairs, 2,000 unique dev images per format. AUC is ranking, not accuracy. Training-encoder metrics; CPU/export checks found zero label disagreements on measured samples with score drift up to 0.0097. GenImage validation confusion matrix (rows actual real/AI, columns predicted real/AI): **[[362, 7], [32, 382]]**.

Baseline and FAILED gate: v0.2.0 to v0.3.0 mean AUC **0.647 to 0.771** original, **0.653 to 0.789** matched; guided matched **0.611 to 0.808**, LDM matched **0.695 to 0.769**. But **as distributed Idm 200 fpr FAILED**: real FPR **22.0% vs 12.8%** (allowed +5 points). Original, stricter-threshold and COCO attempts all failed this check. Selected model accepts this tradeoff. Guided recall at a diagnostic 5% FPR budget: **13.0% to 42.6%**; externally fitted comparison thresholds are not deployed.

Final reserved evaluation (disclosed second use)

GLIDE/DALLE (4,500 images, disclosed second use): macro AUC **0.565 to 0.849** as distributed, **0.643 to 0.833** matched. DALL-E 0.730, GLIDE 0.884-0.895 (as distributed). Accuracy/macro-F1 (as distributed): 74.7% / 0.626. **COCO reals (354, first use):** real FPR 1.69% to **0.28%** as distributed, 11.02% to **1.69%** matched. The 22.0% development-set FPR finding does **not** reproduce on COCO - domain-specific to LAION-style imagery, not universal. No organizer score claimed; AUC is ranking, not accuracy. 2021-2023 vintage generators; organizer's set may differ. Full record: report/releases/v0.3.0/reserved_summary.json.

User diagnostic and reference

Private 11 ChatGPT + 18 phone images, v0.2.0 to CLIP: as-uploaded AI detections **0/11 to 9/11**, real flags **8/18 to 9/18**; matched detections **2/11 to 9/11**, real flags **2/18 to 4/18**. Small-sample CLIP AUC 0.732/0.788. **No accuracy percentage claim from 29 images**; no fitting, per-image detail private.
B-Free reference: 0.970 original / 0.945 matched development AUC on the same images. Third-party weights, not our trained result or deployed component.

Limitations and submission checks

Weaker explanation localisation: only **17/40** audit images satisfy the masking-support rule; the other 23 are disclosed as not localised. Deletion tests p=0.29/0.82. No verified visual-defect or AI-object localisation; human usefulness review pending.
Real-photo false positives remain high; unknown generators/calibration can fail and centre crops omit borders. New-model degradation benchmarks and updated demo pending; screenshot simulation is not device capture. No generator attribution, caption consistency or C2PA signature verification.

Sources: github.com/WisconsinAIVision/UniversalFakeDetect (method/evaluation); github.com/openai/CLIP (backbone); github.com/GenImage-Dataset/GenImage; CIFAKE (Bird & Lotfi). Code and full experiments: github.com/sibtainmunshi/Signal_Scope.
Head SHA-256: 51e1e07d9b0fed13895b7f7a7fe7b207133cae68eca585582fb3c9c05606613f